FIELD TAXONOMY AND SYSTEMATICS

Taxonomy classifies organisms based on shared characteristics, while systematics examines their evolutionary relationssships. In this module, you will explore how these fields help understand biodiversity and trace the ancestry of life on Earth, guiding the identification and classification of species.

Field of Taxonomy and Systematics

Classifying organisms can be likened to categorizing things in life as important and not important. Early human civilization tried to distinguish edible from poisonous plants. It became a way to communicate to other humans which plants are safe to eat.

To date, biologists have identified and named at least 1.5 million species. This number is just small compared to the species that have yet to be discovered. Every year, at least 13,000 new species are identified.

Taxonomy is a science that deals with the classification of organisms based on shared characteristics. It comes from ancient Greek words, *taxis,* which means "arrangement," and *nomia,* which means "method". The group in each level is called a **taxon** (plural: taxa).

For example, for the dog, the taxonomic classification at the order level is *Carnivora*, while Canidae represents the family level, and so on. In addition to these taxonomic names, organisms also have common names that people generally use, such as domestic dogs or wolves. All taxon names are capitalized except for the species, and the genus and species are italicized. When referring to an organism, scientists use both the genus and species names, collectively known as the scientific or Latin name. This system of naming is called ***binomial nomenclature***.

The scientific name of the wolf is *Canis lupus*. Recent DNA studies of domestic dogs and wolves indicate that the domestic dog is a subspecies of the wolf rather than a separate species. As a result, it is assigned an additional name to reflect its subspecies status, *Canis lupus familiaris*.

On the other hand**, systematics** is a broader discipline that seeks to establish evolutionary relationships between organisms, both living and extinct. While taxonomy focuses on classification, systematics places organisms within an evolutionary framework, often incorporating genetic, morphological, and fossil evidence to trace their ancestry and lineage.

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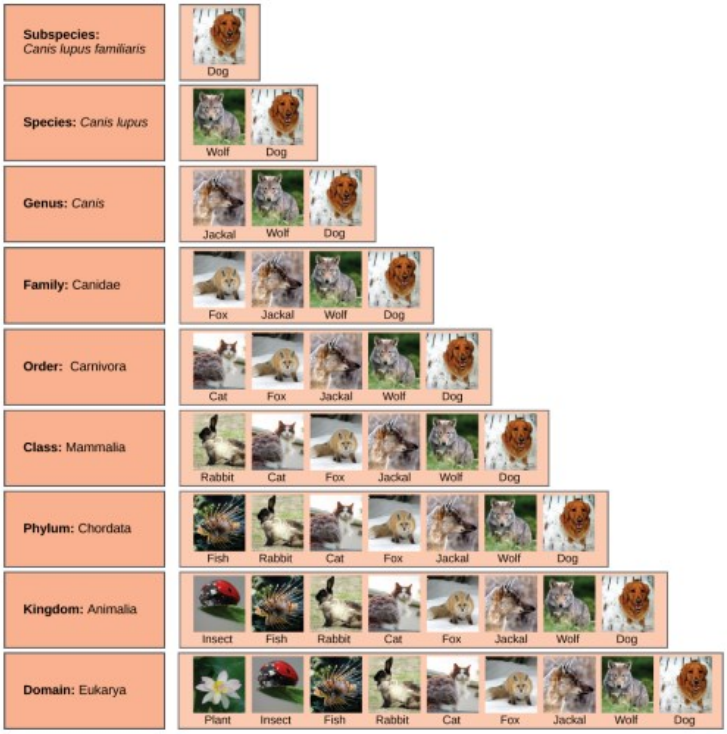


Figure 1. Taxonomic Classification of Domesticated Dogs.
Source: Fowler, S., Rebecca, R., Wise, J. (2023) Concepts of Biology. https://openstax.org/details/books/concepts-biology

Hierarchy of Evolution

The concept of hierarchies in evolution pertains to the organization of biological entities at various levels, each with its own distinct properties and evolutionary processes. This framework is known as the genealogical hierarchy, encompassing the following levels:

Microevolution - refers to small-scale changes within populations over relatively short periods. It involves variations in gene frequencies within a population, often driven by mechanisms like natural selection, genetic drift, mutation, and gene flow. The following levels are involved in microevolution:

1. **Genes** – Mutations and genetic recombination result in variations in genetic material, leading to differences in traits.
2. **Cells** – Genetic changes influence cellular processes, affecting an organism's structure and function.
3. **Organisms** – Natural selection acts on individuals, favoring advantageous traits that enhance survival and reproduction.
4. **Populations** – As genetic variations accumulate, populations evolve, leading to observable changes in traits over generations.

Macroevolution - involves large-scale evolutionary changes that occur over long periods. It focuses on the origin of new species, evolutionary trends, and mass extinctions. The levels within macroevolution include:

1. **Species** – When populations become genetically distinct and reproductively isolated, new species emerge through speciation.
2. **Clades** – Species with a common ancestor form clades, representing branches on the tree of life. Understanding clades helps in studying evolutionary relationships.

3. **Lineages** – Evolutionary lineages trace the ancestry and descent of organisms, often spanning millions of years.

Cladistics, Phenetics and Phylogenetics

There are three major ways in which classifications are reconstructed:

1. **Phylogeny** is the natural system of classification based on the evolutionary history or genealogy shared by a group of organisms. It describes the organism's relationship, such as from which organisms it may have evolved or to which species it is most closely related. This provides information on shared ancestry but not necessarily on how organisms are similar or different.

   Scientists use a tool called a **Phylogenetic tree** to show the evolutionary pathways and connections among organisms. It is known as the *Tree of Life*, which illustrates how different organisms evolved and shows the relationships among them.

   Some trees have a clear starting point, called a *root*, which represents a common ancestor from which all the organisms in the tree evolved. These are called **rooted trees**. At the base of this tree is **LUCA** (**Last Universal Common Ancestor**), which represents the most ancient organism from which all life on Earth descended. From LUCA, the tree branches out into the three major domains of life: **Bacteria**, **Archaea**, and **Eukarya**.

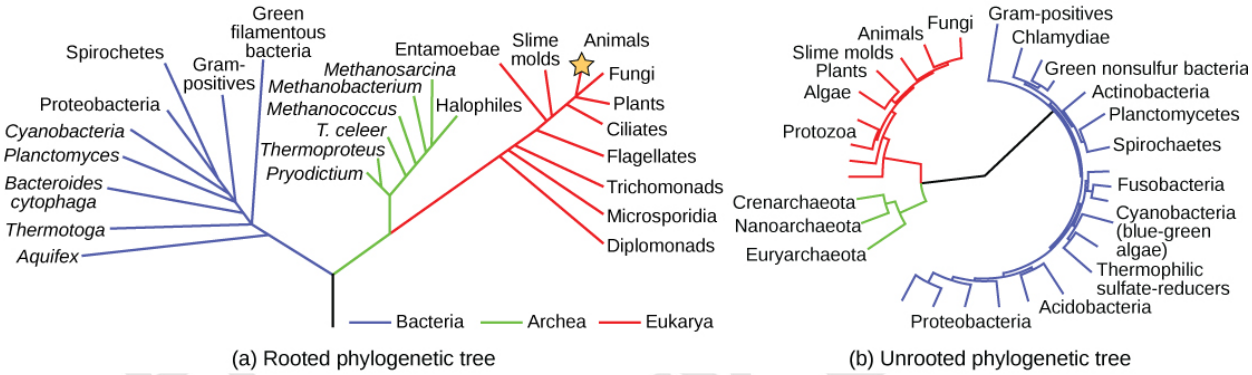


Figure 2. Rooted and unrooted phylogenetic tree

Source: Fowler, S., Rebecca, R., Wise, J. (2023) Concepts of Biology. https://openstax.org/details/books/concepts-biology

   In a rooted tree, the branching indicates evolutionary relationships. The point where a split happens is called the **branch point.** This represents a single lineage that evolved into a distinct new one.

   The phylogeny of organisms is not a permanent classification map because discoveries and data about organisms can change; thus, current phylogenetic trees may be revised as needed.

2. **Cladistics** is the analytical method of refining the evolutionary classification of organisms. This is done by identifying shared derived characteristics. These characteristics are traits that evolved in only one species and are present only in its descendants.

   Shared characteristics are shown in the **cladogram**. It is a diagram that represents an evolutionary relationship between different groups of taxa called "*clades*". This reconstructs the evolutionary history of taxa.

For example, all the animals in the cladogram share a backbone, indicating that they are *vertebrates*. As evolution progresses, new traits appear that differentiate groups of organisms. The presence of hair distinguishes mammals, such as horses, wolves, leopards, and domestic cats, from tortoises. Among the mammals, the wolf, leopard, and domestic cat share a carnivorous diet, setting them apart from the herbivorous horse. Further along the evolutionary line, the leopard and domestic cat possess retractable claws, a feature that enhances their ability to hunt. Finally, the domestic cat is unique in its ability to purr, a trait that distinguishes it even from its close relative, the leopard. This cladogram visually represents how species have evolved, accumulating shared characteristics while also developing unique traits that define their lineage.

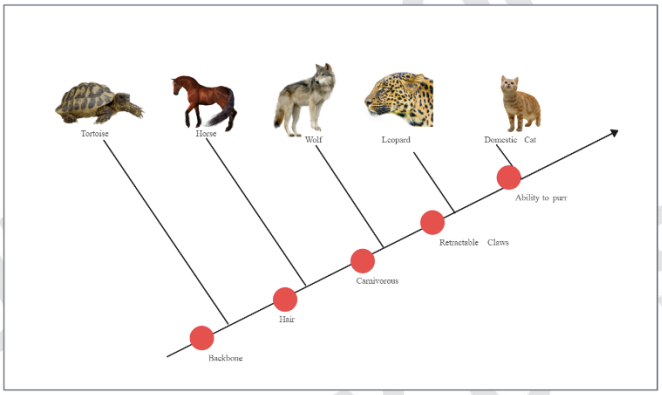


Figure 3. Tortoise to Cat Cladogram
Source: Mihindukulasuriya (2024). https://creately.com/guides/introduction-to-cladograms/

Another way of looking into the cladogram is by noting the lengths of its branches. The length of each branch signifies the length of time by producing a change in characteristics. These lines and their corresponding length can be thought of as timelines that indicate when particular characteristics emerged or when a specific lineage deviated from the common ancestor.

3. **Phenetics**, also known as Numerical Taxonomy, is a classification method that groups organisms based on overall similarity in their characteristics without considering evolutionary relationships. Developed by Sneath and Sokal, this approach treats all characters equally, quantifying the similarity between organisms through a matrix of attributes. It produces phenograms that visualize these relationships, using numerical prefixes to denote similarity levels. While it simplifies classification, phenetics has been criticized for ignoring evolutionary history, which can lead to misleading groupings. Despite this, it remains an operational tool for identifying patterns within diverse species.

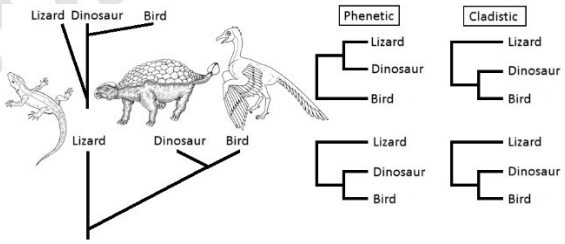


Figure 4. Comparing Phenetic and Cladistic Approaches to Bird Classification
Source: Carr, S. (2024). https://www.mun.ca/biology/scarr/Phenetic_vs_Phylogenetic_classification_of_Birds.html

A **phenogram** is a graphical representation used in phenetics to visually show the relationships between organisms based on their overall similarity. It clusters species according to shared morphological traits, placing those with more similar features closer together in a tree-like diagram. While a phenogram helps identify patterns of physical resemblance, it does not reflect evolutionary ancestry, as it is based purely on observable traits rather than evolutionary history. Therefore, while useful for illustrating general patterns of similarity, a phenogram does not provide insights into how species are related in terms of their evolutionary origins.

Tracing Evolution

Linnaean Classification

A system developed by **Carl Linnaeus** in the 18th century. It is a method of categorizing and naming organisms. It is based on shared characteristics and evolutionary relationships among living things. This hierarchical system helps scientists understand the diversity of life, trace evolutionary origins, and categorize organisms into various levels, from broad to specific. The primary categories in this system are known as the "**taxonomic ranks**," with each kingdom being one of the most basic classifications. Over time, as scientific knowledge has grown, the Linnaean system has been expanded to include more kingdoms, some of which are based on genetic information.

1. Kingdom Archeabacteria

Archaebacteria (or Archaea) are a group of prokaryotic microorganisms that are distinct from eubacteria (true bacteria). Archaea thrive in extreme environments, such as extreme heat, salt, or acidity, and have unique biochemical properties, like different cell membrane structures and metabolic processes. Archaebacteria are classified into several phyla based on their environmental adaptations and metabolic pathways.

A. **Phylum Euryarchaeota**: Key groups include:
   i. **Methanogens**: These organisms produce methane as a metabolic byproduct and are typically found in anaerobic environments like marshes, intestines, and deep-sea vents.
   ii. **Halophiles**: These are salt-loving archaea that thrive in environments with high salinity, such as salt lakes and salt mines.
   iii. **Thermophiles**: Heat-loving organisms that can survive in extremely hot environments, such as hot springs and hydrothermal vents.

B. **Phylum Crenarchaeota**: Members of this phylum are often found in extremely hot or acidic environments, such as hot springs or deep-sea vents. Many are thermophilic (heat-loving) and acidophilic (acid-loving).

2. Kingdom Eubacteria

Eubacteria are also prokaryotic, but they are more common and can be found in various environments. Their classification within phyla is based on factors such as shape, metabolism, and genetic structure.

A. **Phylum Firmicutes**: This phylum includes a wide variety of bacteria, many of which are Gram-positive and have the ability to form endospores, enabling them to survive extreme conditions. Examples include:
   I. **Bacillus:** Soil bacteria, some of which can cause disease but also have applications in biotechnology.
   II. **Clostridium:** Includes pathogens such as *Clostridium botulinum*, which causes botulism, and *Clostridium tetani*, which causes tetanus.

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B. **Phylum Proteobacteria**: A diverse group that includes many important pathogenic bacteria, such as Escherichia coli, Salmonella, and Vibrio species. They are often Gram-negative and have a wide range of metabolic types.
   I. **Escherichia coli**: A common bacterium in the intestines that can be harmful in certain strains.
   II. **Salmonella**: A major cause of food poisoning.
   III. **Vibrio**: Includes Vibrio cholerae, the causative agent of cholera.

C. **Phylum Actinobacteria**: Known for their high G+C content, these bacteria include both beneficial soil bacteria (e.g., Streptomyces) and harmful pathogens (e.g., Mycobacterium).
   I. **Streptomyces**: Soil-dwelling bacteria that produce antibiotics.
   II. **Mycobacterium**: Pathogens responsible for diseases such as tuberculosis and leprosy.

D. **Phylum Cyanobacteria**: Also known as blue-green algae, these bacteria are photosynthetic and play an essential role in producing oxygen. They are found in aquatic environments and contribute to the nitrogen cycle.

3. Kingdom Protista

Protists are a highly diverse group that includes unicellular organisms and some multicellular forms, and their classification is based on their methods of obtaining food and how they move.

A. Phylum Euglenophyta: Euglenoids, like Euglena, are characterized by their ability to photosynthesize or consume other organisms depending on environmental conditions. They have flagella for movement.

B. Phylum Dinoflagellata: Dinoflagellates are mostly marine plankton, some of which are photosynthetic, while others are heterotrophic. Some species cause harmful algal blooms (red tides).

C. Phylum Chlorophyta: Green algae which are similar to land plants in terms of their photosynthetic pigment. Examples include Chlorella and Volvox.

D. Phylum Apicomplexa: This group includes parasitic protists like *Plasmodium*, the causative agent of malaria.

E. Phylum Ciliophora: Ciliates, such as *Paramecium*, use hair-like structures called cilia for movement and feeding. They are often found in freshwater environments.

4. Kingdom Fungi

Fungi are primarily decomposers and are classified based on their reproductive structures and lifestyle.

A. **Phylum Ascomycota**: This is the largest phylum of fungi and includes organisms like yeasts (e.g., *Saccharomyces*) and molds (e.g., *Neurospora*). They reproduce sexually via ascospores.

B. **Phylum Basidiomycota**: Includes mushrooms, puffballs, and other fruiting fungi. They produce basidiospores and play an important role in decomposing organic material.

C. **Phylum Zygomycota**: Includes fast-growing molds such as *Rhizopus* (black bread mold), which reproduce sexually through the formation of zygospores.

D. **Phylum Chytridiomycota**: The chytrids are primarily aquatic fungi, and many are parasitic, like *Batrachochytrium*, which affects amphibians.

5. Kingdom Plantae

Plants are multicellular, photosynthetic organisms, and their classification into phyla depends on their reproductive structures, vascular tissues, and life cycle.

- **Phylum Bryophyta:** Non-vascular plants like mosses, which lack vascular tissues for transporting water and nutrients. They reproduce via spores.

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- **Phylum Pteridophyta:** Ferns and other seedless vascular plants that have vascular tissues and reproduce through spores.
- **Phylum Coniferophyta:** Conifers (like pine trees) are gymnosperms with seeds exposed to cones. They are woody plants that dominate many temperate forests.
- **Phylum Angiosperms (Anthophyta):** Flowering plants that produce seeds enclosed within fruits. This phylum is further divided into monocots (e.g., grasses) and dicots (e.g., roses).

6. Kingdom Animalia

Animals are multicellular, eukaryotic, and heterotrophic. They are classified into phyla based on their body plan, symmetry, and developmental patterns.

- **Phylum Porifera:** Sponges are simple animals with porous bodies and lack true tissues. They are primarily aquatic and filter feed.
- **Phylum Cnidaria:** Includes jellyfish, corals, and sea anemones. These animals have radial symmetry and specialized cells called cnidocytes for stinging.
- **Phylum Arthropoda:** This is the largest and most diverse phylum, encompassing insects, spiders, and crustaceans. Arthropods have segmented bodies, exoskeletons, and jointed limbs.
- **Phylum Chordata:** This phylum includes vertebrates (animals with backbones) like fish, amphibians, reptiles, birds, and mammals, as well as some invertebrates like tunicates and lancelets.

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